GEK-83925

DRIVE SYSTEM INSTRUCTIONS

μdrill™ 3000
RIG DRIVE SYSTEM

DRIVE SYSTEMS OPERATIONS
1501 ROANOKE BOULEVARD
SALEM, VIRGINIA 24153

µdrill™ 3000
RIG DRIVE SYSTEM
INSTRUCTION BOOK

GEK-83925

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The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, Drive Systems Operations, Salem, Virginia, U.S.A.

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µdrill™ is a trademark of General Electric Company

HOW TO USE THIS INSTRUCTION BOOK

This instruction book contains all the necessary information to start up, operate, maintain, and troubleshoot your GE μdrill™ 3000 rig electric control system.

As shown in the Table of Contents in Tab 1, the book is logically arranged according to your needs. The convenient colored tabs will help you quickly find the information you require.

Before using this book, be sure to look over the Table of Contents and read the Introduction in Tab 1.

NOTE, WARNING, AND CAUTION NOTICES USED IN THIS PUBLICATION

WARNING

Warning notices are used in this publication to emphasize that hazardous voltages, currents and temperatures that could cause personal injury exist in this equipment.

CAUTION

Caution notices are used where equipment might be damaged if care is not taken. In situations where inattention could cause either personal injury or damage to equipment, a Warning notice is used.

NOTE

Notes merely call attention to information that is especially significant in understanding and operating the equipment.

REVISION STATUS

If this is a requisition-specific book (that is, if the cover page contains a customer name, requisition number, and rig number), any pages that are revised by General Electric will be forwarded to you for inclusion in this instruction book. Use this page to record such revisions. All revised pages will be marked with a revision notice and date. ***Note that changed pages will be forwarded only to holders of requisition-specific books. All other books carry a revision disclaimer notice on the cover page.***

SECTION	PAGE	REVISION DATE	SECTION	PAGE	REVISION DATE

CHAPTER 1
INTRODUCTION

This manual describes the location, purpose, and operation of the μdrill™ 3000 drive system controls and indicators. The μdrill™ 3000 is a modular rig drive system with individual SCR bridges for each motor. A typical rig drive system includes a driller's console, power control equipment (often mounted in a power control room), a mud pump console (optional), and the necessary motors and generators. Figure 1 is a simplified diagram of a typical μdrill™ 3000 rig drive system.

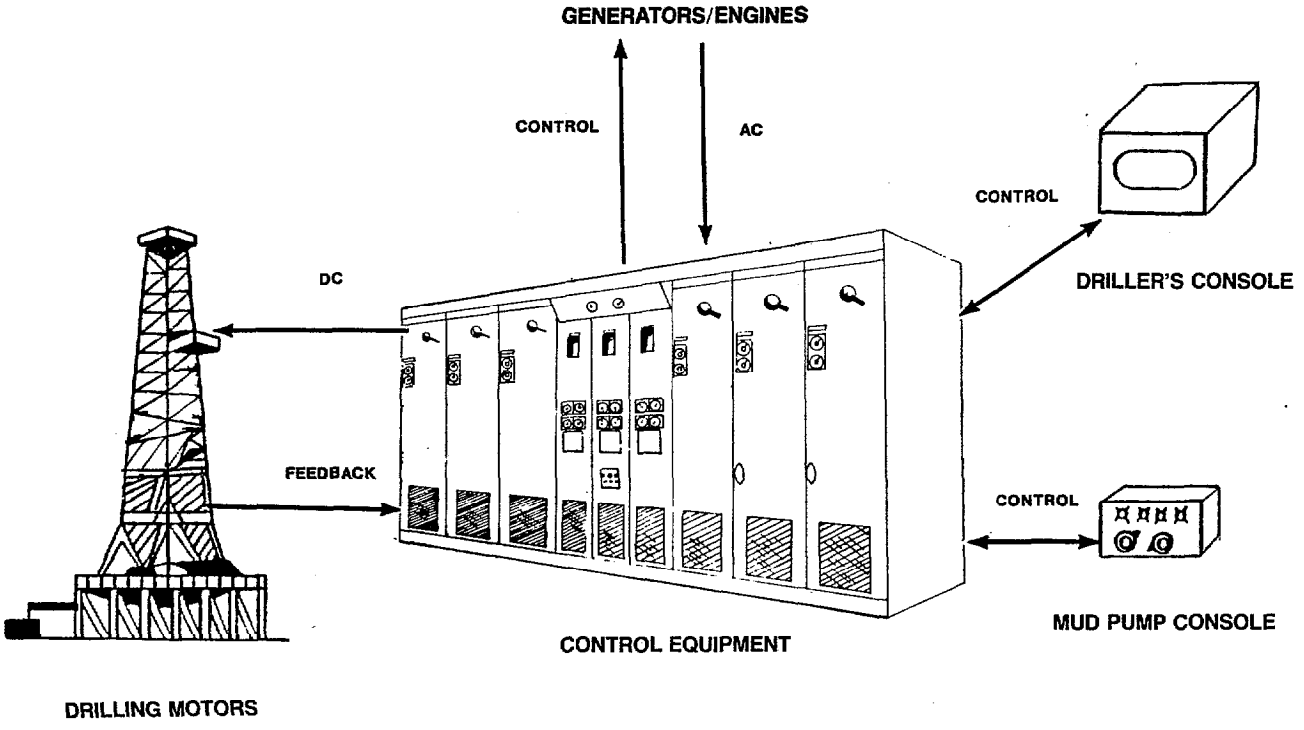


Figure 1.
Typical Rig Drive System

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INTRODUCTION

It is important that you become familiar with the organization of this instruction book so that you can quickly find the information you require for any specific purpose. Below is a summary of the information contained in each of the sections of this book.

SECTION 1 - Table of Contents and Introduction

Contains a table of contents and information on how to use your µdrill 3000 instruction book.

SECTION 2 - How to Care for Your µdrill 3000 Equipment

Contains proper handling and storage procedures for the equipment prior to start-up or during rig moves, as well as information on the acceptable operating environment (temperature, humidity and potentially harmful contaminants). Preventive maintenance procedures are also covered.

SECTION 3 - Rig Up and Spud In Data

Includes a summary of the services provided by the GE Installation and Service Engineering team, and a checklist for customers to record rig-up and spud-in data (such as settings and adjustments) to aid future troubleshooting or tune-up.

SECTION 4 - How to Operate Your µdrill 3000 Equipment

This section is primarily for the drilling operator. It describes the loction, purpose and operation of the µdrill 3000 drive system controls and indicators. The section includes procedures for rig start-up, drilling motor and engine-generator selection, operation and orderly shutdown procedures.

SECTION 5 - How Your µdrill 3000 System Works

The µdrill 3000 control system is made up of highly structured panels interconnected to form a system. This section describes the various panel types and how they are interconnected. It includes a simplified system one-line, various panel configurations, and selected detailed one-lines showing signal flow.

SECTION 6 - How Your AC Generator Control Panel Works

Not Used

SECTION 7 - How Your Feeder Breaker Panel Works

Not Used

SECTION 8 - µdrill 3000 Drilling Operator Stations

Describes the control details of the driller's console and mud pump console. See section 4 of this instruction book for details on how to use these consoles for rig operation.

SECTION 9 - How Your DC Interface Module (XJBF) Works

Describes how the DC interface module distributes control signals beteen the driller's console, motor auxiliary switches, and power conversion panels.

SECTION 10 - How Your DC System Modules Work

Describes the control panels and associated control modules required to convert 600Vac power from the thru bus to variable voltage dc power required to run the rig drilling motors. Includes power conversion, regulator and sequencing functions.

SECTION 11 - How Your Alternate Assignment Panel Works

The μdrill 3000 control system usually provides one power converter panel per drilling motor to eliminate the complications of switching. However, an alternate assignment of a specific motor (i.e., independent rotary table) to a back-up power conversion panel is possible. This section describes the optional alternate assignment panel and its associated devices.

Most sections of this instruction book include a detailed table of contents and introduction for that Section. Familiarize yourself with the details of the apropriate section you need before proceeding with any specific activity (such as operating the equipment, making adjustments, troubleshooting problems, etc.).

Additional device information may be obtained from your Renewal Parts book. This book contains a listing of recommended spares and exploded views of devices (to identify subordinate device components).

SECTION 2
HOW TO CARE FOR YOUR μdrill 3000 EQUIPMENT

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— NOTES —

SECTION 2.1
RECEIVING, HANDLING, AND STORING

RECEIVING THE EQUIPMENT

All equipment is carefully inspected and packed before leaving the factory. Upon receipt, an immediate inspection should be made to identify any missing or damaged items. If damage is evident or if there is visual indication of rough handling, a claim for damage should be filed immediately with the carrier. Both the transportation company and the General Electric Company should be notified. Information as to damaged parts, part number, case number, and requisition number should accompany the claim.

HANDLING THE EQUIPMENT

Specific details such as panel length, width, weight, and lifting provisions are included in the panel outline drawings in Section 12 of this instruction book. The center of gravity of the equipment should always be carefully observed to prevent tipping during handling.

STORING THE EQUIPMENT

Equipment shipped as Power Control Rooms (PCRs) is designed for outdoor storage, provided that all openings are sealed (such as where the air conditioner was removed for shipping). Equipment shipped as panel line ups must be placed immediately under adequate cover; the packing coverings do not protect the equipment for outdoor storage.

In all cases the control equipment should be kept clean, dry, and protected from extreme temperature variations, high humidity and dust. Variations in storage temperature can cause moisture condensation on the equipment. Maintaining the internal temperature of the equipment 5°C to 10°C higher than outside ambient will eliminate this condensation. This may best be accomplished by adding space heaters or using panel space heaters where supplied (this requires external AC supply voltage).

Equipment may be stored in environments of 5% - 95% relative humidity with the above provisions to prevent condensation. Covering material should be breathable (canvas type); do not use plastic. Absolute storage temperature limits are between -40°C and 70°C.

— NOTES —

SECTION 2.2
OPERATING ENVIRONMENT

The following operating environment must be maintained to ensure proper operation and maximum equipment life:

- Operating temperature 0°C to 50°C
- Relative humidity 5% to 95% non condensing
- Altitude 3500 feet maximum without derating
- Corrosive gas content (parts per billion):

Hydrogen chloride	4 PPB
Chlorine	2 PPB
Hydrogen sulfide	2 PPB
Sulphur dioxide	4 PPB
Chlorine dioxide	4 PPB

- Dust: Particle size — 10 - 100 microns
  Concentration — 0.0033 grams/cu. ft. air

NOTE

Effect of corrosive gas and dust is worse at high humidity and can cause electrical short circuits with condensate. For best results, keep equipment cool, dry and clean.

— NOTES —

SECTION 2.3
PREVENTIVE MAINTENANCE

Section 14 of this instruction book contains preventive maintenance advice for specific devices. The following comments are offered as general preventive maintenance recommendations.

WARNING

This equipment contains a potential hazard of electrical shock or burn. Remove all power from the rig drive system before performing preventive maintenance.

INSPECTION ROUTINE

A periodic routine of equipment cleaning and inspection should be established. Such cleaning is necessary to prevent faulty operation due to the accumulation of dust and corrosion. Inspections should be more frequent if demands on the equipment are exacting. Under normal conditions the protective devices do not operate, therefore these devices need careful inspection. If the equipment is in operation while being inspected and a short interruption of service can be permitted, the manual tripping of a protective device will give you an opportunity to check its operation.

LOG BOOK

It is recommended that a regular inspection and maintenance program be followed and a log book be kept. Specific items checked and frequency of inspection varies from rig to rig (to a large part it is a function of environment and housekeeping practices).

MAINTENANCE RECOMMENDATIONS

1. LOOSE CONNECTIONS. Check for loose connections. Vibration may cause terminal board and bus bolt connections to become loose. Loose connections permit moisture and dust accumulations that can result in loss of low level signals at terminal board connections or thermal runaway at bus connections. Problems caused by loose connections can appear and disappear due to vibration.

2. DAMAGED INSULATION. Check for damaged insulation on wires and cables, which can cause intermittent shorting of signal or power cables.

3. DUST ACCUMULATION. Check for dust accumulation on insulation and wire surfaces. Dust accumulating on stand-off insulators can collect enough moisture to produce a current path from bus bars to chassis ground. Dust (especially metallic welding dust) on wire surfaces can cause "tracking" between receptacle pins. Tracking is usually capacitive in nature and involves a build-up of electrical charge along the wire surface, resulting in intermittent problems which are very difficult to locate. Clean bus bars and stand-off insulators with a clean dry cloth (do not use any solvents). Clean wire surfaces between page receptacles by blowing dry air (maximum pressure 80 p.s.i.) across all wire surfaces, directing the air source so that foreign objects are removed rather than relocated.

4. SCR HEAT SINK ASSEMBLIES. Where very dirty conditions exist, there may be enough accumulation of foreign matter to restrict the air flow across the heat sink assemblies. ***Before performing any maintenance in the SCR power bridge, ensure that AC bridge power is locked out.*** To remove dirt from the heat sinks, clean them with a strong industrial vacuum cleaner using a brush attachment. Clean the heat sinks while they are mounted in place, but avoid scratching the bus.

5. PRINTED WIRING BOARDS. It is recommended that a very dirty module be replaced rather than cleaning each printed wire board. If this is not possible, boards can be cleaned in a lukewarm (100°F) solution of mild household detergent or mild solution of a laboratory detergent like ALCONOX or NEUTRA-CLEAN. Use a soft-bristled brush such as a paint brush to scrub the board. After cleaning, rinse thoroughly in several baths of lukewarm, fresh water. Shake off excess water, immerse the board in isopropyl alchohol and agitate, then air dry for a few hours. Do not use solvents containing ammonia, aldehydes, alkalies, aromatic hydrocarbons, or ketones.

6. SHORT CIRCUITS. Normally, the overcurrent protective device on the circuit will prevent any electrical damage except at the actual point of short-circuit. After any large fault current be sure to make a thorough inspection of the entire system to ensure that there has been no mechanical damage to conductors, insulation or equipment. In addition, the individual overcurrent protective device or devices which performed the short-circuit interruption must be inspected for possible arcing damage to contacts, arc chutes and/or insulation.

7. ARCING DAMAGE TO INSULATION. Some organic insulating materials carbonize and lose their insulating qualities when subjected to the heat of an electrical arc. Any insulation found to be carbon-tracked must be replaced before applying power.

8. ALUMINUM BUS CONNECTIONS. Any time aluminim bus connections are being made or tightened, use the following torque values:

   - For a bolt size of 3/8 - 16, torque to 10-15 lb. ft. or finger tighten the nut and then turn 1 1/4 turns more.

   - For a bolt size of 1/2 - 13, torque to 25-35 lb. ft. or finger tighten the nut and then turn 1 1/2 turns more.

9. ALUMINUM BUS REPLACEMENT. Use the following procedure when replacing a piece of aluminum bus:

   a. Clean the plated contact surfaces with a lint free cloth, but ***do not use solvents.*** Coat the contact surfaces with a semi-conductive lubricant such as zinc chromate grease.

   b. Bolt the bus together using a standard coarse thread cap screw (usually 1/2 - 13), two extra thick washers per bolt, a spring lock washer, and a hex nut. Refer to item 8 above for torque values. Do not use Belleville washers as they might scratch the bus surface.

   c. After the joint has been bolted, brush a thin coat of grease (zinc chromate) over the joint area. Wipe off excess with a lint free cloth. Do not use a solvent. Do not get grease on supports or insulators.

   d. Plated aluminum can be bolted to other plated aluminum, or to plated or unplated copper. Do not bolt to unplated aluminum.

10. DC DRILLING MOTORS. To ensure maximum performance of your DC drive system, the following motor checks should be made:

a. Motor Cooling

- Air filters (if used) should be cleaned or replaced periodically.
- Any duct work should account for a 12% reduction in motor cooling for each 90% duct bend.

  NOTE: Both air filters and duct work should be planned before a motor blower is selected. These devices usually cause reductions in motor air flow.
- If heat exchangers are used, special motor blower impellors and larger motors blowers (twice the horsepower of an open air flow system) are usually recommended by the motor manufacturer.

b. Motor Brushes

Sticking motor brushes cause severe motor arcing, and do not carry their share of the armature current. Check all motor brushes for wear and freedom of movement at least once a month or after 500 hours of operation.

c. Motor Cleanliness.

Moisture, dirt, paint, and bugs are the usual enemies of good motor performance. These enemies cause brushes to stick, air flow to be retarded, and commutator bars to short. Using the motor manufacturer's recommended techniques, perform motor cleaning routines at least once a month or after 500 hours of operation. For specific information about motor maintenance, motor cooling, and motor operation ask your motor supplier for details. The Transportation Systems Business Division of General Electric (Erie, Pennsylvania) builds the GE-752 DC motor, and they have many specific publications that concern these subjects.

— NOTES —

SECTION 3
RIG STARTUP AND SERVICE

A start-up inspection is performed on all GE control systems. The inspection assures you the GE equipment meets the purchase specifications. During the course of a start-up operating personnel become familiar with the equipment. An on-site school will be conducted to further their understanding of the control. If any GE equipment fails, it will be identified and replaced under warranty.

The start-up inspection will be conducted by I&SE, the Installation and Service Engineering Division of General Electric. I&SE maintains service representatives trained on GE equipment in all parts of the world.

If you have a problem and are unable to contact a local I&SE representative, please call your service manager in Salem, VA:

Dennis Bogh
(703) 387-8321
(703) 389-1452
TLX: 82-9337

Please contact the local I&SE office at least two weeks before you plan to start diesel engines or energize equipment. The I&SE representative will arrange a pre-start-up meeting with you to discuss the start-up schedule and responsibilities. Careful coordination between I&SE and your crew will be necessary to avoid delays. Following is an outline of a standard I&SE start-up inspection:

A. Collect nameplate data
   1. Motors
   2. Generators
   3. Transformers

B. Perform mechanical inspection
   1. Check all contactors, relays, and breakers for smooth operation.
   2. Check structure and welds for signs of damage.
   3. Check that equipment is clean, dry, and free of foreign materials.
   4. Inspect GE motors and generators.

C. Perform electrical inspection
   1. Megger all motors, generators, blowers, AC buswork, breakers, and main transformers.
   2. Check governor power supply.
   3. Check connections of engine speed sensor and actuator.
   4. Check that MCC heater sizing is correct per elementary.

D. After engines are started:
   1. Energize generators.
   2. Check phase rotation.
   3. Check synchronizing circuits.
   4. Adjust governor gain and stability.
   5. Adjust voltage regulator gain and stability.

E. Energize motor control centers

F. Energize DC power converters and controls
    1. Inspect electrical operation of each power converter.
    2. Verify current and speed limits.
    3. Verify the KW limit and KVA limit.
    4. Inspect control circuit operation.

G. Operate motors
    1. Bump motors for rotation.
    2. Inspect motor operation at full speed.
    3. Measure and record vibration.
    4. Inspect equipment operation under load if load is available.

It will be the customer's responsibility to operate the equipment. I&SE will rectify all warranty claims during the start-up.

I&SE does offer spud-in inspection. The purpose of a spud-in inspection is to assure the GE equipment operates as specified under load. The best time for this inspection is one to two weeks after the rig begins to drill. At least 80% load of one engine must be available to allow dynamic adjustments. During spud-in inspection, engines are tested to verify their ability to carry rated load. KW and KVA limits are verified under dynamic conditions. Motors are inspected while operating to assure proper commutation. Any warranty problems that occur during this time will be corrected with no additional charge to the customer.

A well trained drilling crew can significantly reduce rig downtime. Training will improve their ability to identify equipment problems and operating problems, allowing proper service to be obtained before a crisis occurs. A two day on-site school will be presented during each start-up. The school will include:

A. Driller orientation to console

B. Explanation of generator synchronizing procedure for motor men and pushers

C. Maintenance orientation for electricians.

This school is conducted for available personnel. If normal operations personnel are not available during startup, special arrangements must be made for training.

Cooperation will be necessary to ensure a safe start-up. It can be hazardous to energize electrical devices unless all parties understand the danger involved at each step. The following are two safety hazards that must be avoided:

1. Working on high voltage circuits alone. A helper must be assigned to the GE representative anytime he works on the high voltage.

2. Making any connections to the PCR during start-up tests. The rig must be completely wired before start-up can begin. Customer personnel will not be allowed to make connections to the SCR equipment under test.

The impact of safety considerations can be significantly reduced by work planning during the pre-start-up meeting.

Modifications, equipment updates, and service contracts are all available from I&SE. Necessary changes to GE equipment are often determined after it ships from DSD, Salem, VA. Most changes can be easily accomplished in the field before or during equipment start-up.

All I&SE work is guaranteed. Equipment modifications made by I&SE will not affect the system warranty.

Two types of service contracts are offered, emergency and maintenance. After the warranty period, I&SE will offer an emergency service contract to continue warranty style service on any rig. I&SE can also provide routine maintenance on a scheduled basis quarterly or semiannually. This maintenance assures more trouble-free operation of rig electrical equipment. All after-sales service will be provided by the local I&SE office. Please contact them with any questions you have.

INSTALLATION & SERVICE ENGINEERING DIVISION

ENGINEERING STANDARDS *installation*

TITLE	REV	DATE	NUMBER
OIL WELL DRILLING RIG EQUIPMENT WITH THYRISTOR POWER CONVERTERS SPLIT START-UP MICRO-DRILL 3000	1	5/82	ES-060-2.5
	ISSUED BY: ELEC. ENGRG.		
	AUTHORIZED BY: ENGRG. & PRGMS. DEPT.		

FOR GE EMPLOYEE USE	FOSSIL		NUC	MAR	INDUSTRIAL			
	GAS	STM	NUC	MAR	PWR	COMP	D&C	MT
							X	

CUSTOMER-USER _____

CUSTOMER-OEM _____

CONTACTS:

NAME	TITLE	PHONE	TELEX

RIG NAME & NO. _____

RIG UP YARD _____

LOCATION _____

REQ. NO. _____

FSR NO. _____

DATE OF REPORT _____

DATE INSTALLATION

BEGAN _____

COMPLETED _____

DATE OF:

SPUD-IN _____

ACCEPTANCE _____

RIG TYPE _____

INITIAL LOCATION _____

ENGINEER _____

GENERAL: The purpose of this Installation Standard is to provide a checklist of the tests and inspections to be performed, and to provide a record of the inspection data. The performance of this service is authorized through the Specialist, Drilling Rigs, Installation and Service Engineering, Salem, VA.

SCOPE: This Standard is specifically applicable to Micro-Drill 3000 drilling rigs utilizing thyristor power conversion, manufactured by GE.

I&SE PROCEDURES

1. Notify DSD Product Service representative as soon as start-up date is known.
2. The Start-Up section of this Standard defines the minimum procedures to be followed during the start-up at a Micro-Drill 3000 drive.
3. The Load-Test section defines load tests to be accomplished if load is available during start-up, or at spud-in.
4. When start-up is complete, the following action must be taken concerning the prints:

I&SE PROCEDURES (Cont'd)

4. (Cont'd)
   a. Mark one set of prints and leave with customer.
   b. Mark a second set of prints and send changes to T. Mitchell, DSD Product Service, Salem, VA.
   c. Forward a complete set of corrected prints to I&SE office in the area where the rig will drill.
   d. To mark prints: red = additions; blue = deletion; green = notes
5. The sequence of start-up events shall be determined during a pre-start-up meeting with all parties concerned.
6. This procedure is designed for a "standard" rig. If the equipment exceeds the charts in this procedure, attach additional sheets.
7. Make note, on the DSD Product Service copy, of any readjustments required on items which should have been set at the factory; particularly on SEM card daughter boards, current limits, and reference.
8. Adjustments will be made accurate to ± 2 percent. kW and kVA limits will be +0, – 10 percent.
9. These instructions are not a substitute for familiarity with the equipment or the instruction book. Be familiar with each test and its expected results before energizing the equipment.

Special Tools:

10. The following instruments, or equivalent, are normally required for the check-out and tune-up procedures associated with this test procedure:
    a. Megger, 0-500 Vdc
    b. Simpson 260 VOM
    c. Fluke 8000A/AS Digital Multimeter (optional)
    d. IRD 320 Vibration Monitor
    e. Phase Sequence Indicator Knopp K-3, 600V
    f. Strobe Tachometer (series rigs only)

WARNING: Comply with all applicable safety precautions when working on this equipment. Special precautions are to be observed due to the large short circuit capacity of this type of equipment.

START-UP:

I. Factory Set Functions:

1. Generator ammeter
2. Suppression
3. F.L. dropout and pickup
4. Gen. module O.V. trip
5. Motor starter heater sizes (verify — often ac motor sizes are changed after shipment)
6. Torquing bolts (except at shipping splits on put-togethers)

These functions are not checked at start-up unless there is reason to suspect they are misadjusted.

II. AC Generator Section Start-up

The ac bus must be energized before any major tests can be done. Before the ac bus is energized, all connections should be visually checked for shipping damage and torque. Any damage or loose connections should be recorded on page 5 or 12.

All equipment must be meggered before it is energized. Record all nameplate data and megger readings on the installation checklist.

Make preliminary adjustments on all governors before engines are started. Start each engine-generator set individually, adjust idle, rated speed, stability and gain. Adjust rated voltage. Check phase rotation. Record data on pages 6 and 7. Check UV trip and UFOV functions. Check bus voltmeter accuracy.

Check synchroscope and sync lights. Use a voltmeter at the PT primaries to verify all phases are correct with the sync scope. Parallel all generators in all combinations and test reverse power function.

This completes the initial ac start-up. Record data on pages 5 and 6.

III. DC Section Start-up

The dc section start-up can be accomplished only after all controls and cables are connected. It is a power test only. The power converters are all tested using internal test capabilities. I/O's are checked by turning on assignments and watching the results. The dc motors are first bumped and then operated.

Record data in section 70.2.

IV. kW and kVA Limit Adjustments

kW and kVA limits both act by reducing the individual drive current limits (ISLIM). They can be statically adjusted by injecting test signals into the GEM module until the module meters read limits, then adjusting the appropriate bias in the kW/kVA module until ISLIM in each drive is reduced to 200 amps. This procedure has a dynamic accuracy of +0, −10 percent for kW and kVA limits. It can be checked under load conditions.

V. Customer Training

Customer training must be held for every Salem rig. The training can be held informally several hours at a time, or formally in a classroom. Agreement should be reached with the customer before the start-up begins.

The training will be (16) hours and should cover enough information to familiarize rig personnel with the GE equipment. The troubleshooting guide provides a good reference on which to base training.

VI. Load Tests

Load tests are done to assure the power converter system will meet all ratings under load. These tests actively check generator load sharing, motor load sharing, kW limits and kVA limits. Limits should be set during start-up and checked during load tests.

Load tests must be done at start-up if load is available. Otherwise, they should be done at spud-in.

PRE-INSTALLATION CHECKLIST

		*YES	NO	OK
20.0	POWER CONTROL ROOM (PRE-POWER)			
20.1	**PCR Ground Circuits**			
	Inspected all skids for grounding based on applicable codes.	☐	☐	☐
20.2	**Visual Inspection**			
	1. Relays and contactors operate freely by hand.	☐	☐	☐
	2. Screwed/bolted connections are tight.	☐	☐	☐
	3. Weld joints are free of cracks.	☐	☐	☐
	4. Assure structure is free of damage.	☐	☐	☐
	5. Meter faces are free from cracks.	☐	☐	☐
	6. PCR and compartment doors open and close easily.	☐	☐	☐
	7. Equipment is free of dust, dirt, excess moisture, oxidation and foreign materials.	☐	☐	☐
	8. Space heaters are connected to proper voltage per elementary drawing.	☐	☐	☐
	9. Arc chute tie-downs and clips removed from relays.	☐	☐	☐
20.3	**Governor Power Supply Checks**			
	Measure battery volts = 12v ± 1 volt. If battery is low recharge as necessary.	☐	☐	☐
20.4	**Modifications**			
	Complete any factory retrofits and customer changes. Briefly, list any incomplete items at time start-up began.	☐	☐	☐

_____

_____

_____

_____

_____

_____

_____

_____

_____

* **Yes-No-OK System:** If equipment does not meet standards, mark "No" and correct the problem. When satisfied equipment is up to standards, mark "OK."

30.0 AC GENERATOR SECTION

30.1 Nameplate Data

	Gen. 1	Gen. 2	Gen.3	Gen. 4
Engine model no.				
Engine mfgr.				
Hp rating				
Rated rpm				
Gen. mfgr.				
Model no.				
Serial no.				
Amps				
kW				
kVA				
Space heater volts				
Exciter stator rated volts				
Exciter stator rated amps				
Actuator model number				

30.2 Pre-Energization Checks

Megger				
Exciter field to ground (Note 1)				
Rotor field to ground (Note 2)				
Armature to ground				
Measure at GEM (Note 3)				
Mag pickup ohms				
Actuator ohms				
Exciter stator field ohms F1, F2				
GEM Serial No. (behind door)				
Megger Gen.				
Phase to ground				
A				
B				
C				
Phase to Phase (Note 4)				

	YES	NO	OK
30.3 Checked that GEM transformers are connected for proper output voltage. Use high tap if generator field exciter is rated more than 35 full load volts. Use low tap if generator field is rated more than 7 amps or less than 35 full load volts. See GEM manual for transformer connections. If generator field requires more than 70 volts, reconnect the field for lower voltage.	☐	☐	☐

Note 1 Disconnect leads at GEM and megger field from leads at GEM at 500V to ground.

Note 2 Jumper diode plates together and megger plates to ground at 500V.

Note 3 Lift leads at GEM and measure from leads at GEM end.

Note 4 Lift generator armature leads at generator. This is to assure multiple conductor generator armature cables connected properly.

40.0 TRANSFORMER(S)

40.1 Nameplate Data

	T1	T2	T3	T4
1. Transformer name				
2. Mfgr.				
3. Model no.				
4. Serial no.				
5. kVA rating				
6. Prim. volt/wdg. conn.	/	/	/	/
7. Sec. volt/wdg. conn.	/	/	/	/

40.2 Visual Inspection

	YES	NO	OK
1. Cables run to appropriate machines. (Gen. 1 to plug 1, etc.)	☐	☐	☐
2. Screw and bolted connections are tight.	☐	☐	☐
3. Weld joints are free from cracks.	☐	☐	☐
4. Inspect for damage.	☐	☐	☐
5. Equipment free from dust, dirt, excessive moisture, oxidation, and foreign materials.	☐	☐	☐
6. Electrical clearances meet applicable codes.	☐	☐	☐
7. All units connected for same transformer taps.	☐	☐	☐
8. Inspected all plug connectors for proper construction.	☐	☐	☐

40.3 Pre-Energization Checks

A. Windings meggered and above minimum requirements (2 megohm at 40°C) using a 500 volt insulation tester.	☐	☐	☐
B. Neutral grounded on Y connected transformers.	☐	☐	☐

45.0 MOTOR CONTROL CENTER

45.1 Visual Inspection

1. Structure free from damage.	☐	☐	☐
2. All compartment doors open and close easily.	☐	☐	☐
3. Meter faces are free from cracks.	☐	☐	☐
4. Equipment is free from dust, dirt, excess moisture, oxidation, and foreign materials.	☐	☐	☐
5. Painted finish is free from noticeable mars.	☐	☐	☐

6. Checked that MCC overload heaters are appropriate for customer auxiliary motors. Checked that motors are per elementary.

Note inconsistencies:

	Motor Name	Hp	Heater
1.			
2.			
3.			
4.			

50.0 ENGINE AND GENERATOR RUNNING CHECKS

50.1 Governor Tests

A. Preliminary Adjustments

1. Adjusted "Idle Speed" potentiometer for rated idle engine speed of 600 RPM or as required by customer at _____ RPM.	☐	☐	☐
2. Adjusted "Rated Speed" potentiometer for rated engine speed of _____ RPM.	☐	☐	☐

50.1 Governor Tests (Cont'd)

	YES	NO	OK
3. Adjusted "Gain" and "Stability" potentiometers per General Electric Module Instruction Book, until stable. Stability pot will be about 10, gain pot about 60 in most cases.	☐	☐	☐
B. Made vibration checks and recorded — Item 85.0 on pg. 12.	☐	☐	☐
C. Checked that magnetic pickup voltage is between 5 and 40 Vac.	☐	☐	☐
D. Adjusted incoming line-to-line generator volts for 600 volts ac at rated engine speed using the pot 105.	☐	☐	☐
E. Checked panel voltmeter, all phases.	☐	☐	☐

50.2 Interconnection Checks — While engine is operating at full speed, no load

A. Checked generator phase-to-phase volt unbalance less than 60 volts.	☐	☐	☐
B. Checked line to ground volts balanced within 35 volts.	☐	☐	☐
C. Checked for proper phase sequence at the line side of the generator control breaker.	☐	☐	☐

50.3 Overspeed/Actuator Adjustments

1. Mechanical overspeed was set to trip at _____ RPM on each prime mover. (Check with customer.)
2. Recorded volts at governor actuator output for rated RPM and No Load.

	GEN. 1	GEN. 2	GEN. 3	GEN. 4
Actuator volts at rated speed (0.8 to 1.8 should be similar ± 0.2 volts)				

50.45 Transformer Energization Checks

A. Checked phase sequence 1-2-3 (line and load side).				☐	☐	☐
B. Checked with at least 10 percent load record.				☐	☐	☐
	T1	T2	T3	T4		
1. Primary Volts						
2. Secondary Volts (Should be within 2 percent of nominal.)						
C. No signs of overheating (discoloration, smoke, smell, etc.).				☐	☐	☐
D. Checked for normal noise levels while recording values listed in Step B.				☐	☐	☐

50.5 Protective Devices

A. Checked long time setting set at 1.0 to 1.15 times generator rated arm amps. (Check instantaneous set at minimum, not below 2 times.)	☐	☐	☐
B. Checked that all breaker interlocking schemes are functional per elementary drawings.	☐	☐	☐
C. Checked Overvoltage/Underfrequency function to verify that underfrequency protection is operating. Switch engine to idle. Check armature volts begin to drop as frequency dips below 55 Hz.	☐	☐	☐

50.6 Synchronizing Equipment Checks — YES / NO / OK

A. Preliminary

1. Checked each generator unit by measuring the voltage deviation between running and incoming phase voltages on each phase. (The synchroscope should be at 12 o'clock and sync lights should not be illuminated when the phase voltage deviation is minimum.) ☐ ☐ ☐

2. Checked running and incoming frequency meters for _____ Hz with the engines at rated speed and the generators at rated no load voltage output. ☐ ☐ ☐

B. Synchronizing Checks

At light load conditions, paralleled all combinations of generators. Verified that sync check relays are adjusted to 5 electrical degrees (+5°, −0°) of synchroscope 12 o'clock position. (Sync check adjustments should be made in accordance with applicable publications.) ☐ ☐ ☐

50.7 Collect GEM data in section 76C.

65.0 DC MOTORS

65.1 Preliminary Checks

	DWA	DWB	MP1	MP2
A. DC Motors				
Mfgr.	_____	_____	_____	_____
Model no.	_____	_____	_____	_____
Serial no.	_____	_____	_____	_____
B. Blower Motor				
Mfgr.	_____	_____	_____	_____
Hp	_____	_____	_____	_____
Rpm	_____	_____	_____	_____
C. Insulation Tests — DC Motors (Note 1)				
Arm to Fld	_____	_____	_____	_____
Arm to Gnd	_____	_____	_____	_____
Fld to Gnd	_____	_____	_____	_____
D. Insulation Tests — Blower Motors (Note 2)				
Phase to Gnd	_____	_____	_____	_____

65.2 Visual Inspection

1. Screwed/bolted connections are tight. ☐ ☐ ☐
2. Checked motors and blowers free of damage. ☐ ☐ ☐
3. Checked for foreign materials inside motor and fan assembly. ☐ ☐ ☐
4. Checked for signs of oxidation and moisture. ☐ ☐ ☐

65.3 Pre-Power Checks

A. Verified with customer that all power and control connections are per elementary diagram. ☐ ☐ ☐

B. Checked that space heaters are connected to proper voltage per elementary diagram and are operational. ☐ ☐ ☐

Note 1 Megger from contactor panel (2 megohms min. for new GE motor; motor will operate with megohm).

Note 2 Megger from MCC (2 megohms min.).

		YES	NO	OK
65.3	**Pre-Power Checks (cont'd)**			
	C. Brushes are free in brush holders and spring pressure is applied.	☐	☐	☐
68.0	**PCP Test**			
	A. Energized all PCPs. Checked that they all pass self test.	☐	☐	☐
	B. Checked all personality plugs correct per elementary.	☐	☐	☐
69.0	**Static Calibration kW Limit**			
	A. Removed plug JCX from the back plane of a generator module (when not in use and turned off). Connected the test signal from the kW/kVA module (JTEST), to TP9 on the HFPA card. Adjusted signal until the module kW meter read the rated kW of one engine.	☐	☐	☐
	B. Adjusted the kW bias on the kW/kVA module until the KVLIM lights (engineering test D) indicate 200 amps on all PCPs. (Watch large time lag; wait 10 sec. for limit to settle.)	☐	☐	☐
	C. Checked that the percent kW meter reads to one engine mark with scale input. Adjusted as necessary.	☐	☐	☐
	D. Readjusted signal at TP9, HFPA until the kW meter read about 90 percent of kW limit. Checked that the high load light came on at this time.	☐	☐	☐
	E. Removed signal and replaced JCX.	☐	☐	☐

kW limit set at _____ kW. Bias _____ volts.

NOTE: Set kW limit to match engine horsepower unless customer requests lower limit.
kW limit = Engine hp × .746 × .95.
Engine hp is on nameplate. A 399 cat may be 1000 to 1300 hp. 0.746 converts hp to kW. 0.95 the approximate generator efficiency.

Example: For a 1215 hp engine set kW limit at:
1215 × .746 × .95 = 860 kW

69.5 Static kVA Limit Calibration

A. Calculate kVA limit signal expected from GEM at full load:

$$V_{kVA} = \frac{-1 \text{ volt}}{200 \text{ amp}} \times \text{Rated gen kVA} \times .96 \text{ or}$$

$$V_{kVA} = \frac{-1 \text{ volt}}{200 \text{ amp}} \times \text{Rated gen amps.}$$

B. Remove wire DkVA from JDC-4 on one generator module which is not in use.

C. Adjust the test jack voltage on the kW/kVA module to V_{kVA} calculated above. This voltage is negative.

D. Input V_{kVA} from the kW/kVA module into wire DkVA removed above.

E. Adjust the kVA bias in the kW/kVA module until KVLIM is 200 amps on all drives [KVLIM = Engineer Test, TWS: D]

F. Re-adjust the signal source to about 90% V_{kVA}. Check the high load light comes on at this time.

G. Remove signal source, reconnect wire DkVA to JDC-4.

H. KVA limit set at _____ amps.
Bias _____ volts.

69.6 Field Heating Adjustment. Repeat for each generator module.

A. With generator running at idle speed, measure V_{R1}, the voltage across R1 on the generator module back plane. (R1 is 0.1 ohm, 10 w. V_{R1} is dc volts less than 3.0 volts.)

B. Obtain the maximum generator field current rating, if rated, from the generator name plate.

C. Calculate $V_{TP10} = \frac{51}{\text{If rated}} = V_{R1}$

Adjust HFPA, TP10 to this value, using pot R113.

	G1	G2	G3	G4
V_{R1}				
If rated				
V_{TP10}				

		YES	NO	OK
70.1	**Assign each motor.**			
A.	Checked all auxiliaries operate correctly.	☐	☐	☐
	Bumped motors for rotation forward.	☐	☐	☐
	Bumped drawworks (DWKS) and rotary motors for rotation reverse.	☐	☐	☐
B.	(Shunt only) Adjusted field current to proper value. R32 on field exciter 752 motor field should be:	☐	☐	☐
	752 field 39 amps at 0 armature current for motor with 5 HP blower. 752 field 44 amps at 0 armature current for motor with 10 HP blower. 761 field 32 amps at 0 armature current for motor with any blower.			
	Field amps can be measured at field exciter TP10, and TP1 (com). 1 volt = 10 amps.			
	Minimum field current is about 80% rated.			
	Checked field volts at motor. 752 motor will have 70 or 78 volts. 761 motor will have about 53 volts.	☐	☐	☐
	Recorded values in sections 70.2 and 70.3.	☐	☐	☐
70.15	**Assign motors to alternate assignments.**			
	Checked auxiliaries operate correctly.	☐	☐	☐
	Checked fields for proper current record in sections 70.2 and 70.3	☐	☐	☐
	Bumped motors for rotation forward.	☐	☐	☐
	Bumped motor for rotation reverse.	☐	☐	☐
70.2	**Assigned DWKS motors. Recorded DW data.**	☐	☐	☐

		DWA	DWB	DWC (or alternate)	ROTARY
	Field current (I_a = 0)				
	REF (max)				
(Note 1)	CLLIM (rt unclutched/dw unclutched)				
	CLLIM (rt clutched/dw unclutched/ check rt variable with torque limit)				
	CLLIM (rt unclutched/dw clutched)				
	Maximum volts (VM)				
	Maximum speed (series only)				
(Note 2)	VGAIN				

Note 1: CLLIM is engineering test E.

Note 2: VGAIN is normal test C.

			YES	NO	OK
Checked DWKS (rotary) recalibration for single and multiple motors on rotary table. Rotary clutch only engaged.	per motor CL max at CLLIM 1 motor _____ 2 motor _____ 3 motor _____				
Checked DWKS recalibration for single and multiple motors assigned to DWKS. DWKS clutch only engaged.	DW CL per motor 1 motor _____ 2 motor _____ 3 motor _____				

Check regeneration operation

1.	Operated DWKS unclutched at full throttle both forward and reverse.	☐	☐	☐
2.	Checked regeneration cycle. Observed panel ammeter regeneration current limit (300A). DWKS clutch must be cycled for regeneration.	☐	☐	☐

3. Recorded data.

	DWA	DWB	DWC
Regen. amps	_____	_____	_____

70.3 Recorded MP data

		MP1A	MP1B	MP2A	MP2B
	REF (max)	_____	_____	_____	_____
	ISLIM (two motors)	_____	_____	_____	_____
	ISLIM (one motor)	_____	_____	_____	_____
(Note 1)	REF (max auxiliary console)	_____	_____	_____	_____
(Note 2)	VGAIN	_____	_____	_____	_____

76. Checked Optional Auxiliaries

A.	Super charger auto start	☐	☐	☐
B.	Mud pump stroke counter	☐	☐	☐

C. Recorded Gen. Module Data	GEN. 1	GEN. 2	GEN. 3	GEN. 4
TSW6	_____	_____	_____	_____
TSW7	_____	_____	_____	_____
Volt. reg.				
P100	_____	_____	_____	_____
P101	_____	_____	_____	_____
P114	_____	_____	_____	_____
Gov. reg.				
P103	_____	_____	_____	_____
P104	_____	_____	_____	_____
P105	_____	_____	_____	_____
P102	_____	_____	_____	_____

Seal all pots adjusted with RTV.

Note 1: ISLIM is engineering test 6.

Note 2: VGAIN is normal test C.

INSTALLATION CHECKLIST

		YES	NO	OK
80.0	ENGINE-GENERATOR LOAD TEST (PART II AT SPUD-IN)			
80.1	Loaded engine with generator by operating mud pumps in a line choke. Minimum 80 percent load on one engine.	☐	☐	☐
	1. Checked engine for constant speed (frequency) and generator for constant voltage (± 3 percent) as load is increased.	☐	☐	☐
	2. Paralleled units and checked stability and load sharing at as high a load as possible.	☐	☐	☐
	3. Recorded following adjustment data.			

	ENG 1	ENG 2	ENG 3	ENG 4
	_____	_____	_____	_____
	_____	_____	_____	_____
Low idle RPM	_____	_____	_____	_____
Actuator Volts, no load	_____	_____	_____	_____
Actuator Volts, full load	_____	_____	_____	_____
Governor power supply volts	_____			

	Speed adjust pots on panel door were in mid position at 60 Hz.	☐	☐	☐
80.2	Applied enough load to reach rated kW load on the engine. Checked that kW is limited before the engine is bogged down. Recorded maximum kW.	☐	☐	☐
80.3	Applied enough DWKS load by using high gear or the brake to reach the specified generator kVA. Checked that the kVA is limited to this value. Recorded max kVA.	☐	☐	☐
80.4	Checked that high limit light on console was energized at the proper time.	☐	☐	☐

Max kW (observed) _____ kW bias _____
Max kVA (observed) _____ (amps) kVA bias _____

80.5	**Complete any items left open at start-up**	☐	☐	☐

85.0 VIBRATION CHECKS

85.1.1 Complete the following table. Maximum acceptable vibration at 1200 RPM is 3 mils. All readings Displacement (mils).

		Gen 1	Gen 2	Gen 3	Gen 4	DWA	DWB	MP 1	MP 2
Drive End	1. Horizontal								
	2. Vertical								
	3. Axial								
Opposite Drive End	4. Horizontal								
	5. Vertical								
	6. Axial								

Comments

90.0 LOAD TEST — YES NO OK

90.1 Operate mud pumps:

	PUMP 1 A	PUMP 1 B	PUMP 2 A	PUMP 2 B
SCR amps				
Max volts (VM)				
Pump strokes per minute				
Pump head (psi)				
Max. speed				

Checked motor temp. OK after 60 min. operation. ☐ ☐ ☐

90.2 DWKS Load Test

A. Observed DWKS operation

	DWA	DWB
Maximum current		
Maximum volts		
Regen. current		

B. Observed DWKS load sharing within 10%. ☐ ☐ ☐

C. Checked motor temperature. ☐ ☐ ☐

100.0 COMPLETION

Marked any adjustment changes in the adjustment data section on page 12. ☐ ☐ ☐

Prints were modified and returned with a copy of this Standard to Salem, DSD. ☐ ☐ ☐

100.1 CUSTOMER TRAINING

Oil well drilling equipment is complex. Customer personnel need special training to understand its operation. In order to assure customer understanding, two days of I&SE time is sold with each rig for this training.

I&SE is charged with the responsibility of training any and all available customer personnel for two days. The training can take several forms depending on the customer availability and desires.

Generally, the customer will have electricians, tool pushers and drillers to be trained. The tool pushers and drillers need to know the following:

1. How the drill console operates.
2. How to synchronize generators.
3. Where to turn on ac equipment.
4. How to "read" the lights on the SCR package.
5. How to reset breakers and the location of internal breakers.
6. What assignments to make for the basic drilling functions, top hole, reaming, tripping, etc.
7. How to do basic troubleshooting with the operator's manual.

The electricians need to know the above and the following:

8. How to read the GE print system.
9. How to troubleshoot the system using the advanced techniques in the operator's manual.

Usually, it is most convenient to give about four hours of lecture and twelve hours of hands-on experience with the rig. The four hours of classroom should be taught in a hotel suite or other classroom away from the rig. (If you stay close to the rig there will be constant interruptions.)

Following is an outline for two days of training covering the basics.

TRAINING OUTLINE

LECTURE OUTLINE

			Date & Time Complete
20 Min.	A.	Introduction 1. Your name and phone number. 2. Introduction of class members and experience levels. 3. Collect list of attendees (use form on page 17).	/ / M
10 Min.	B.	Description of Electric Rig 1. Qualities. 2. ac section and ac bus. 3. ac power conversion. a. Simple controls. b. Assignments. 4. ac auxiliaries.	/ / M
30 Min.	C.	AC Generator Section 1. Qualities. 2. ac section and ac bus. 3. dc power conversion. a. Simple controls. b. Assignments. 4. ac auxiliaries.	/ / M
30 Min.	D.	Mud Pump Assignment 1. Pump — SCR combinations available. 2. Possible pump problems.	/ / M
45 Min.	E	DWKS and Rotary Assignments 1. How to set up drawworks for tripping. 2. How to set up rotary table for drilling. 3. Rotary current limit. 4. Spinout function.	/ / M
15 Min.	**Break**		
90 Min.	F.	Basic Troubleshooting from Operator's Manual	
30 Min.	G.	General Maintenance 1. dc motor. 2. ac generator. 3. ac induction motors. 4. Electronics and PCR house. (KEEP IT CLEAN AND DRY!)	/ / M

MOVE CLASS TO RIG

Date & Time Complete

120 Min. A. AC Generators

1. Show each individual how to start one generator.
2. Show each individual how to parallel.
3. Show each individual how to turn generators off.

/ / M

60 Min. I. Drill Console

1. Explain each knob.
2. Let drillers make assignments.
3. Explain regeneration function.

/ / M

60 Min. J. Troubleshooting

Put some simple problems in rig and ask rig personnel to locate problems:

1. Turn off blower starter.
2. Unplug auxiliary motor.

/ / M

DISMISS ALL NON-ELECTRICAL PERSONNEL

3 Hrs. K. Print Reading

1. Explain how to read GE prints.
2. Explain rig prints functionally.

/ / M

5 Hrs. L. Troubleshooting for Electricians.

Generally, if a rig is available, the best training for electricians is locating "bugs" placed in the system:

1. Remove personality plug from SEM.
2. Lift throttle wire at drill console.
3. Lift wire from actuator.
4. Short out contactor auxiliary to turn on alarm horn.
5. Lift contactor control wire.

/ / M

You should answer questions as they come up, but allow the electricians to troubleshoot the problem themselves. The operator's manual has a good troubleshooting guide that electricians should be encouraged to use.

GENERAL ELECTRIC
OIL WELL DRILLING
LAND RIG EQUIPMENT
CUSTOMER TRAINING

Date: _____ **Instructor:** _____

Location: _____ **Address:** _____

_____ **Phone** _____

Attendees:

Name	Occupation	Company